10.5 Effect Size

Section 9.5 ("Potential Hypothesis Testing Pitfalls and Ethical Issues") discusses the issue of the practical significance of a statistically significant test and explains that when a very large sample is selected, a statistically significant result can be of limited importance. To illustrate this principle, suppose that a web designer wanted to compare the mean download time (in seconds) for two different web page designs without making the assumption of equal variances in the respective populations. Figure 10.14 presents the results of this comparison based on a sample of 50 users of each design.

FIGURE 10.14
Excel separate-variance *t* test results for the difference in the download time between users of page design A and page design B for samples of 50 users of page design A and 50 users of page design B

	A	B
1	**Separate-Variances *t* Test for the Difference Between Two Means**	
2	(assumes unequal population variances)	
3	**Data**	
4	Hypothesized Difference	0
5	Level of Significance	0.05
6	**Population 1 Sample**	
7	Sample Size	50
8	Sample Mean	0.96
9	Sample Standard Deviation	0.2000
10	**Population 2 Sample**	
11	Sample Size	50
12	Sample Mean	0.9
13	Sample Standard Deviation	0.1700
14		
15	Intermediate Calculations	
16	Numerator of Degrees of Freedom	0.0000
17	Denominator of Degrees of Freedom	0.0000
18	Total Degrees of Freedom	95.5208
19	Degrees of Freedom	95
20	Standard Error	0.0371
21	Difference in Sample Means	0.0600
22	Separate-Variance *t* Test Statistic	1.6163
23		
24	**Two-Tail Test**	
25	Lower Critical Value	-1.9853
26	Upper Critical Value	1.9853
27	*p*-Value	0.1093
28	Do not reject the null hypothesis	

The results indicate a p-value of $0.1093 > \alpha = 0.05$, so at the 0.05 level of significance, you would conclude that there is no evidence of a difference in the mean download time -between the two designs.

Now, suppose that instead of taking a sample of 50 users of each design, the web designer examined the results for 5,000 users of each design. Figure 10.15 presents the results of this second comparison.

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FIGURE 10.15
Excel pooled-variance *t* test results for the difference in the download time between users of page design A and page design B for samples of 5,000 users of page design A and 5,000 users of page design B

	A	B
1	**Pooled-Variance *t* Test for the Difference Between Two Means**	
2	(assumes equal population variances)	
3	**Data**	
4	Hypothesized Difference	0
5	Level of Significance	0.05
6	**Population 1 Sample**	
7	Sample Size	5000
8	Sample Mean	0.96
9	Sample Standard Deviation	0.2
10	**Population 2 Sample**	
11	Sample Size	5000
12	Sample Mean	0.9
13	Sample Standard Deviation	0.17
14		
15	Intermediate Calculations	
16	Population 1 Sample Degrees of Freedom	4999
17	Population 2 Sample Degrees of Freedom	4999
18	Total Degrees of Freedom	9998
19	Pooled Variance	0.0345
20	Standard Error	0.0037
21	Difference in Sample Means	0.0600
22	*t* Test Statistic	16.1632
23		
24	**Two-Tail Test**	
25	Lower Critical Value	-1.9602
26	Upper Critical Value	1.9602
27	*p*-Value	0.0000
28	Reject the null hypothesis	

With such large sample sizes, the assumption of equal variances can be ignored and the pooled–variance *t* test and the separate-variance *t* test will provide nearly identical results; therefore, either test can be used. Now, because of the larger sample size, the *p*-value is $0.0000 < 0.05$, and you would conclude that there is evidence of a difference in the mean download time between the two designs. With even larger sample sizes, such as those that are commonly used when analyzing big data, the likelihood of finding a statistically significant but perhaps practically unimportant difference between the means of two groups *increases*.

Therefore, finding a way to measure the magnitude of a result other than its statistical significance and *p*-value becomes imperative. One alternative method is to measure the **effect size**. A simple way to measure the effect size is to construct a confidence interval estimate of the difference between the two population means using Equation (10.2) on page 313 and determine the width of the confidence interval. These confidence interval results for the download time example with samples of 5,000 in each of the two groups are shown in Figure 10.16.

FIGURE 10.16
Excel results for the 95% confidence interval estimate of the difference in the mean download time between users of page design A and page design B for samples of 5,000 users of page design A and of 5,000 users of page design B

	D	E
3	**Confidence Interval Estimate**	
4	**for the Difference Between Two Means**	
5		
6	**Data**	
7	Confidence Level	95%
8		
9	Intermediate Calculations	
10	Degrees of Freedom	9998
11	*t* Value	1.9602
12	Interval Half Width	0.0073
13		
14	**Confidence Interval**	
15	Interval Lower Limit	0.0527
16	Interval Upper Limit	0.0673

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With these results, you conclude with 95% confidence that the difference in the mean download time between users of page design A and page design B is between 0.0527 seconds and 0.0673 seconds. However, you do not know whether the magnitude of the effect size is small, medium, or large. Bonett (reference 1) developed a standardized effect size statistic shown in Equation (10.8) and a confidence interval estimate of the effect size shown in Equation (10.9) (Bonnett). Research by Cohen suggests that a standardized effect size of 0.2 can be classified as small, a standardized effect size of 0.5 can be classified as medium, and a standardized effect size of 0.8 or more can be classified as large (Cohen).

CONFIDENCE INTERVAL ESTIMATE FOR BONETT'S STANDARDIZED EFFECT SIZE (ES_B) FOR THE MEAN DIFFERENCE IN TWO POPULATIONS

$$ES_B = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2 + S_2^2}{2}}} \tag{10.8}$$

$$\frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2 + S_2^2}{2}}} \pm Z_{\alpha/2}\sqrt{\frac{ES_B^2(S_1^4 + S_2^4)}{8\left(\frac{S_1^2 + S_2^2}{2}\right)^2(n_m - 1)} + \frac{2}{(n_m - 1)}} \leq E \leq \tag{10.9}$$

$$\frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2 + S_2^2}{2}}} - Z_{\alpha/2}\sqrt{\frac{ES_B^2(S_1^4 + S_2^4)}{8\left(\frac{S_1^2 + S_2^2}{2}\right)^2(n_m - 1)} + \frac{2}{(n_m - 1)}} \leq E \leq$$

$$\frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2 + S_2^2}{2}}} + Z_{\alpha/2}\sqrt{\frac{ES_B^2(S_1^4 + S_2^4)}{8\left(\frac{S_1^2 + S_2^2}{2}\right)^2(n_m - 1)} + \frac{2}{(n_m - 1)}}$$

where

ES_B = sample estimate of Bonett's Standardized Effect Size

E = population value of Bonett's Standardized Effect Size

$Z_{\alpha/2}$ = upper tail critical value of the standardized normal distribution

S_1 = sample standard deviation from population 1

S_2 = sample standard deviation from population 2

n_M = mean sample size where $n_M = \frac{n_1 + n_2}{2}$

To demonstrate the computation of Bonett's standardized effect size in Equations (10.8) and (10.9), return to the example concerning the difference in the download time between 5,000 users of page design A and 5,000 users of page design B. Figure 10.17 shows Excel worksheet results for the effect size for this example.

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FIGURE 10.17
Excel worksheet for Bonett's standardized effect size and 95% confidence interval estimate of Bonett's standardized effect size of the difference in the mean download time between users of page design A and page design B for samples of 5,000 users of page design A and of 5,000 users of page design B

	A	B
1	**Bonnett's Standardized Effect Size**	
2		
3	**Data**	
4	Confidence level	95%
5	**Population 1 Sample**	
6	Sample Size	5000
7	Sample Mean	0.96
8	Sample Standard Deviation	0.2
9	**Population 2 Sample**	
10	Sample Size	5000
11	Sample Mean	0.9
12	Sample Standard Deviation	0.17
13		
14	Intermediate Calculations	
15	Difference between Means	0.0600
16	Mean of Variances	0.0345
17	Square Root of Mean of Variances	0.1856
18	Bonett's Effect Size	0.3233
19	First Numerator	0.0003
20	Mean Sample Size minus 1	4999
21	First Denominator	1377.7244
22	Standard Error	0.0200
23	*Z* Value	-1.9600
24	Half Width of Interval	0.0392
25		
26	**Confidence Interval**	
27	Interval Lower Limit	0.2841
28	Interval Upper Limit	0.3625

Using Equation (10.8),

$$ES_B = \frac{0.96 - 0.90}{\sqrt{\frac{(0.20)^2 + (0.17)^2}{2}}} = 0.3233$$

and using Equation (10.9)

$$0.323263 \pm (1.96)\sqrt{\frac{(0.3233)^2((0.20)^4 + (0.17)^4)}{8\left(\frac{(0.20)^2 + (0.17)^2}{2}\right)^2(5{,}000 - 1)} + \frac{2}{(5{,}000 - 1)}}$$

$$0.2841 \leq E \leq 0.3625$$

Thus, you can conclude with 95% confidence that Bonett's standardized effect size is between 0.2841 and 0.3625. According to Cohen's criteria, this effect size would be classified between small and medium, but closer to small than to medium.

Effect Size for the Difference Between Two Proportions

Bonett's standardized effect size statistic measured the effect size for the difference between two population means. There are several alternative measures available to measure the difference between two population proportions (the Berenson, Grissom, and the two Fleiss references). However, a simple way to measure the effect size is to construct a confidence interval estimate of the difference between the two population proportions using Equation (10.6) on page 334 and determine the width of the confidence interval.

To illustrate this method, suppose that Web designers tested a new call to action button on its web page. Every visitor to the web page was randomly shown either the original call to action button (the control) or the new call to action button. The metric used to measure success was the download rate: the number of people who downloaded the file divided by the number of people who saw that particular call to action button.

Of 4,883 visitors to the web page who saw the original call to action button, 421 downloaded the file. Of 5,698 visitors to the web page who saw the new call to action button, 787 downloaded the file. Figure 10.18 presents the confidence interval estimate calculated from these data.

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FIGURE 10.18
Excel results for the 95% confidence interval estimate of the difference in the proportion of visitors to the web page who downloaded the file using the original call to action button and the new call to action button

	D	E
3	**Confidence Interval Estimate**	
4	**of the Difference Between Two Proportions**	
5		
6	**Data**	
7	Confidence Level	95%
8		
9	Intermediate Calculations	
10	Z Value	-1.9600
11	Std. Error of the Diff. Between Two Proportions	0.0061
12	Interval Half Width	0.0119
13		
14	**Confidence Interval**	
15	Interval Lower Limit	-0.0638
16	Interval Upper Limit	-0.0400

With these results, you conclude that the difference between the original call to action button and the new call to action button in the proportion of visitors who would download the file is between −0.0638 and −0.0400. In other words, the new call to action button has an effect size of between 0.0400 and 0.0638 as compared to the original call to action button.

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EG10.5 EFFECT SIZE

Use the **COMPUTE worksheet** of the **Effect Size workbook** as a template.

The worksheet already contains the data for the Figure 10.17 example concerning the difference in the download time between 5,000 users of page design A and 5,000 users of page design. For other problems, change the confidence level and the sample statistics in cell range **B4:B12**. Open to the COMPUTE_FORMULAS worksheet to examine the details of the COMPUTE worksheet formulas.

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